

## Trained



train



Large labelled  
training set

Corpus-specialized  
features

A

Standard



Cross-env.



## Training-free, prior work



Frozen patch  
features



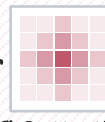
Codebook

or



Segmentation CLS attention

or



Prior-selected / aggregated features

B

Standard



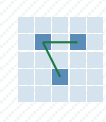
Cross-env.



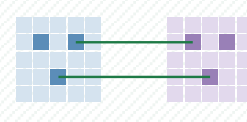
## Ours



Frozen patch  
features



Within-image  
Balanced, matchable patch evidence



Across-image  
Balanced, matchable patch evidence

C

Standard

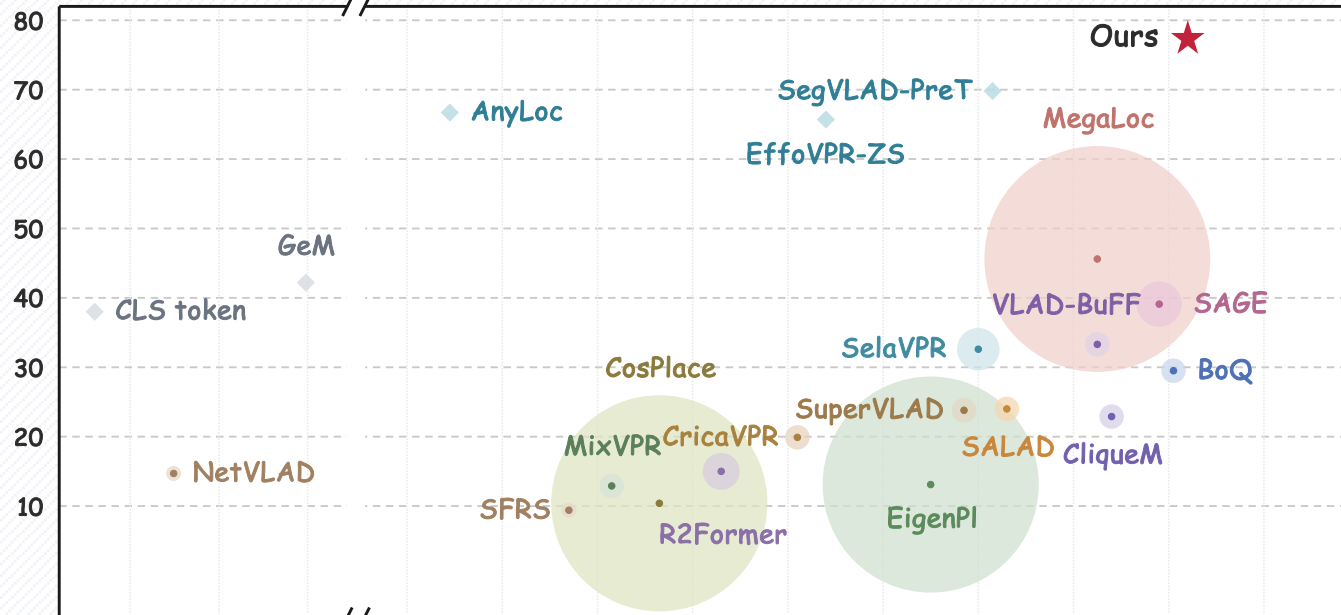


Cross-env.



## Training cost vs. performance

VP-Air R@1 (cross-environment benchmark, %)



Tokyo 24/7 R@1 (standard benchmark, %)

D

★ ours (training-free)

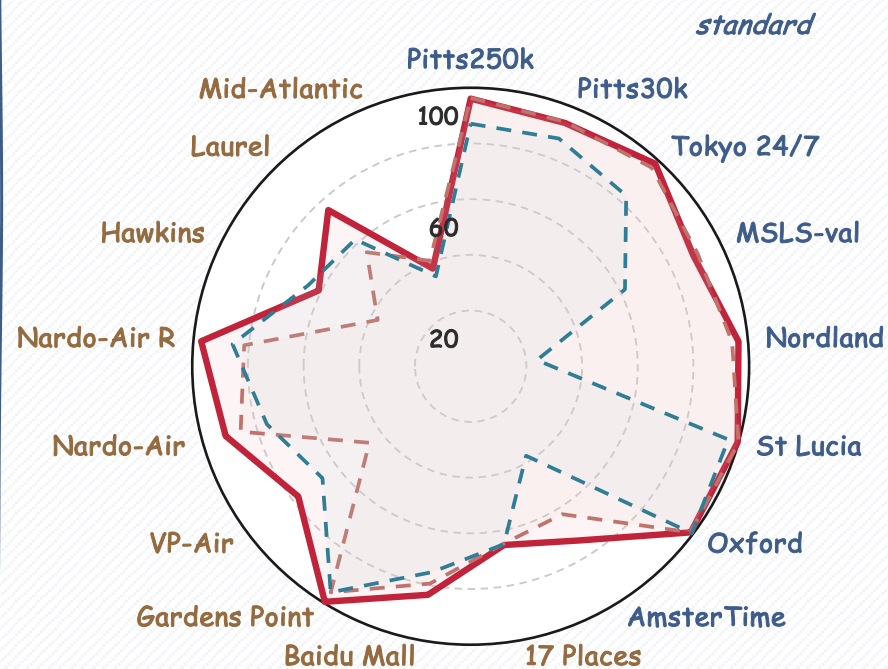
♦ training-free

● trained

● 1 million images

● 5 million images

## Performance across 17 benchmarks



cross-environment

E

— Ours

— MegaLoc

— AnyLoc